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Dual-comb spectroscopy over a 100 km open-air path

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Dual-comb spectroscopy (DCS) provides broadband, high-resolution, high-sensitivity amplitude and phase spectra within a short measurement time, thus holding promises for atmospheric spectroscopy. However, previous research has been limited to measuring over open-air paths of about 20 km. Here, by developing a bistatic set-up using time–frequency dissemination and high-power optical frequency combs, we implement DCS over a 113 km turbulent horizontal open-air path. We successfully measure the absorbance spectra of CO_2 and H_2O with a 7 nm spectral bandwidth and a 10 kHz frequency accuracy, and achieve a sensing precision of <2 ppm in 5 min and <0.6 ppm in 36 min for CO_2. We anticipate our system to find immediate applications in the monitoring of urban greenhouse gas and gaseous pollutants emission. Our technology may also be extended to satellite-based DCS for greenhouse gas monitoring and calibration measurements.

Atmospheric spectroscopy is a key technique used to study the chemical and physical properties of the Earth's atmosphere by analysing the interaction of light with atmospheric molecules and particles. This technique, especially when it is satellite-based, has become a valuable tool in the study of atmospheric composition[1–4], structure[5–7] and dynamics[8–11], which has essential applications in global climate change, carbon budget assessment, deep space exploration and air pollution research.

So far, atmospheric spectroscopy techniques such as grating spectrometers[12–14], integrated path differential absorption lidar (IPDA lidar)[15], heterodyne spectroradiometers[16,17] and Fourier transform spectrometers (FTS)[18] have been able to provide remote ground-based or satellite-based gas spectroscopy[19–26] with various temporal and spatial resolutions and sensitivities.

Among these methods, grating spectrometers, FTS and heterodyne spectroradiometers require sunlight and cannot work at night. IPDA lidar, typically using a narrow-linewidth laser, has difficulty measuring multiple gas species. Open-path tunable diode laser absorption spectroscopy (TDLAS) is an incoherent method, and its sensing distance is limited, which makes it difficult for long path urban monitoring and satellite-based applications[27,28]. To realize global coverage, current satellite-based instruments are usually in low orbit, resulting in a long revisit time (for example, OCO-2 for CO_2 measurements has a revisiting time of 16 days)[29–31]. New methods with fast and

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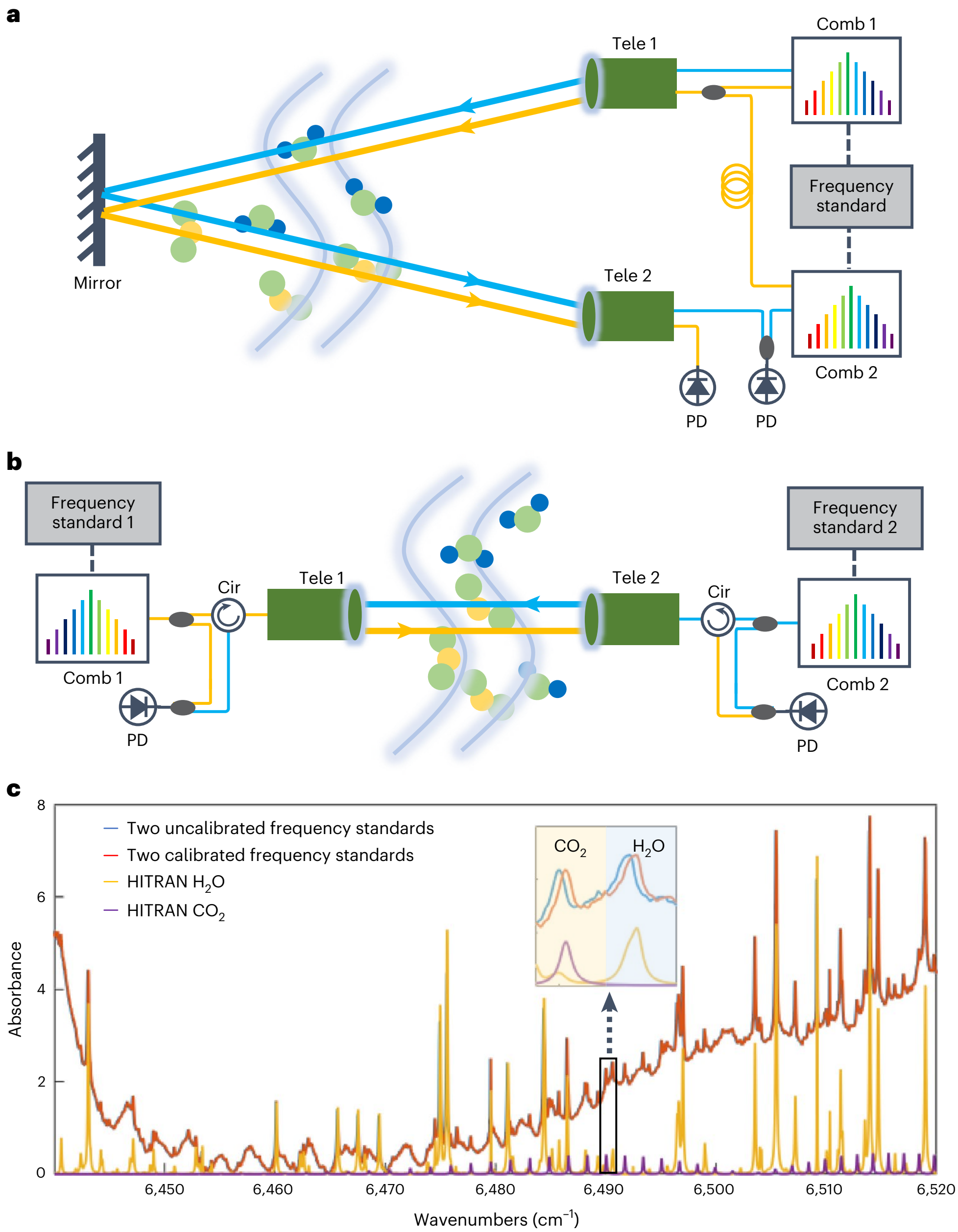

Fig. 1 | Protocols of DCS. a, Two classic configurations used in the monostatic DCS protocol: dual combs interfere before passing through the atmosphere (orange lines) and one comb first passes through the atmosphere and then interferes with the local comb (blue lines). The molecules in the figure refer to GHGs typically measured in the open atmosphere, such as CO_2 and H_2O. Tele, telescope; PD, photodetector. **b**, Our bistatic two-way protocol: two combs separated into two remote terminals are sent to each other through the atmosphere and interfere with each other at the receiving terminals. Cir, circulator. **c**, Comparison of measured spectra with HITRAN. Two independent frequency standards result in a frequency shift of about 1.1 GHz without frequency difference calibration (blue line). The two frequency standards are calibrated using time–frequency transfer and eliminate the frequency offset (red line).

continuous measurements are needed to fulfil the study of localized point carbon sources and sinks in high temporal–spatial resolution. Recently, open-air dual-comb spectroscopy (DCS)[32–40], as a novel coherent spectroscopic tool, has proven to be a superior candidate technology for precise, accurate and continuous fast multi-gas measurements. DCS can cover a broadband of 1–10 µm, allowing the realization of active remote sensing of large numbers of gas species using only a single dual-comb system.

Since the first demonstration, numerous DCS experiments have been spawned. The technology has been exploited in oil field monitoring[32], urban vehicular greenhouse gas (GHG) emission[37], livestock emission measurement[35], urban GHG monitoring[33] and so on. Owing to high acquisition speed for a single interferogram, the intensity noise and piston-like phase noise caused by turbulence are frozen[34,36]. Therefore, DCS has shown excellent immunity to turbulent speckle. Coherence between each interferogram makes coherent averaging possible, which can resist background noise such as shot noise and detector noise. The advantages of immunity to turbulent speckle and background noise result in the capability of sensing longer distances without calibration, which is considered to be a perfect precision spectroscopy tool for remote sensing of the atmosphere. Meanwhile, DCS has other advantages of multiple comb mode with absolute

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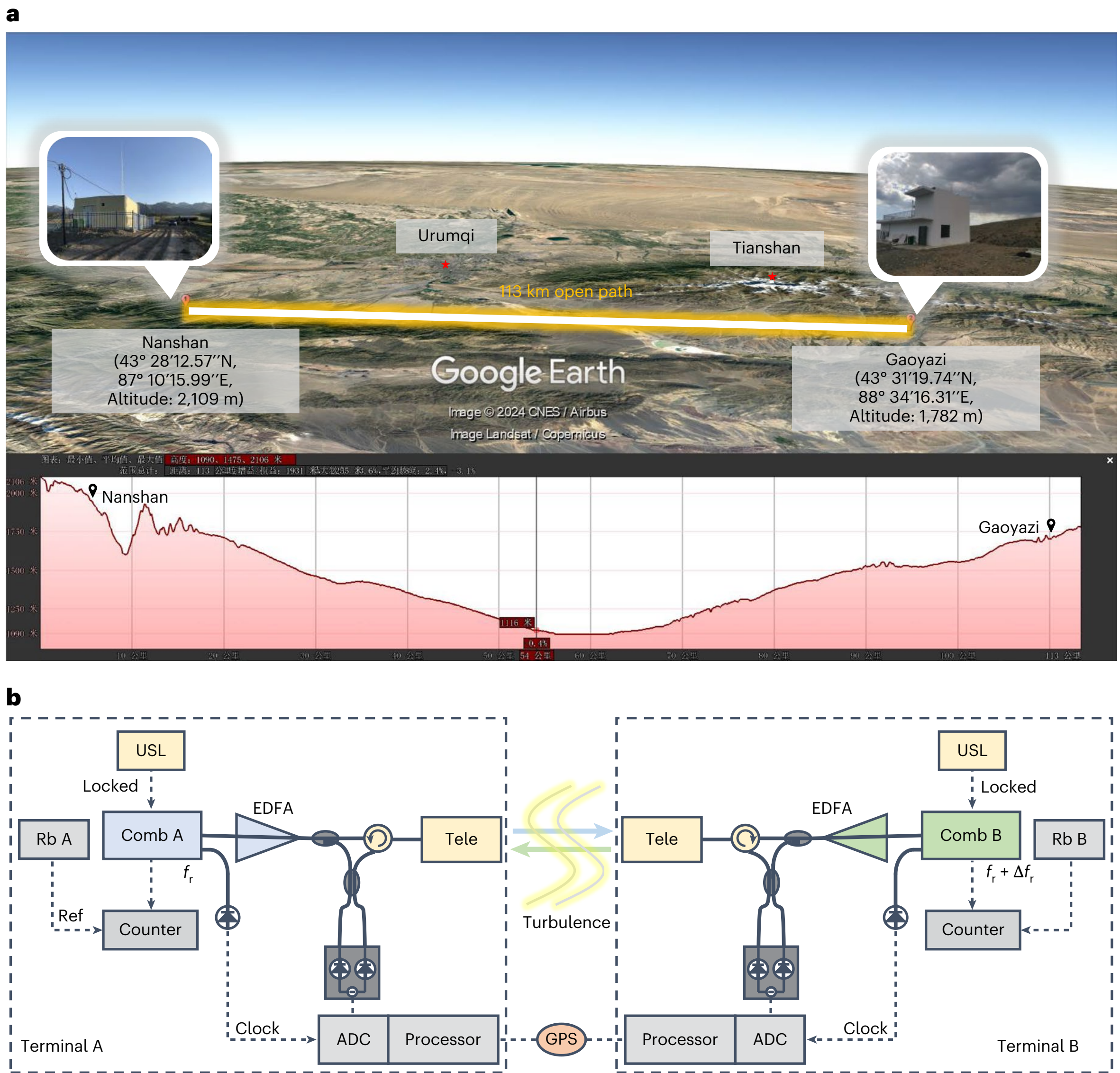

Fig. 2 | The experimental set-up. a, Overview of the 113 km DCS. Map data: Google Earth, 2022 Maxar Technologies, Landsat/Copernicus. **b**, The main configuration and set-up at each terminal. Rb, rubidium clock; USL, ultrastable laser; EDFA, erbium-doped fibre amplifier; Tele, telescope; ADC, analogue-to-digital converter; GPS, Global Positioning System.

accuracy, stability with negligible linewidth and high-brightness coherent light sources. Furtherly, by developing networks consisting of satellite-ground links and horizontal links, the spatial distribution and variations of GHG could be acquired, contributing importantly to global climate change research[41]. By establishing a DCS in geosynchronous orbit, it is possible to monitor fast-changing sources, which can be a worthy supplement to the existing GHG satellites. Compared with existing GHG satellites, geosynchronous DCS has a limit to global coverage, but it will be possible to monitor point carbon sources and sinks continuously and precisely by the combination of GEO stations and long horizontal ground-to-ground DCS paths in the future[42]. In the meantime, it can be a useful tool to calibrate the sun-synchronous orbit satellite GHG measurements with high accuracy in day and night, not being affected by the solar zenith angles as the TCCON be[43].

However, the longest reported distance for DCS is limited to less than 20 km (refs. 33,44), with a link loss of 40–60 dB. This is lower than the typical channel loss of 60–80 dB for high-orbit satellite-to-ground links using single-mode fibre coupling[45–47]. Meanwhile, as will be discussed later, the existing configuration of the DCS experiment with folded path measurement proves impractical for the satellite-to-ground link owing to the severe geometric loss exceeding 100 dB. Thus, the demonstration of DCS over long distances with high channel loss in the horizontal open air becomes inevitable and urgently required for satellite-based applications[48]. In addition, hard underlying surface in urban could lead to degradation of the urban surface thermal environment, which will bring negative effects on urban air quality, plant growth and water environment[49]. DCS measurements over a 100 km horizontal range can cover any city, allowing long-term monitoring of changes in water vapour and carbon dioxide levels. This will provide insight into the impact of urban land surfaces and the effects of urbanization on the environment[50,51].

Set-up

The main obstacle to the longer distance of DCS is the low signal-to-noise ratio. We try to increase the signal-to-noise ratio on both the protocol and hardware sides. Figure 1a shows the current monostatic DCS protocol[52] used in the vast majority of the open-path DCS experiments. Two optical combs are locked to the same terminal by

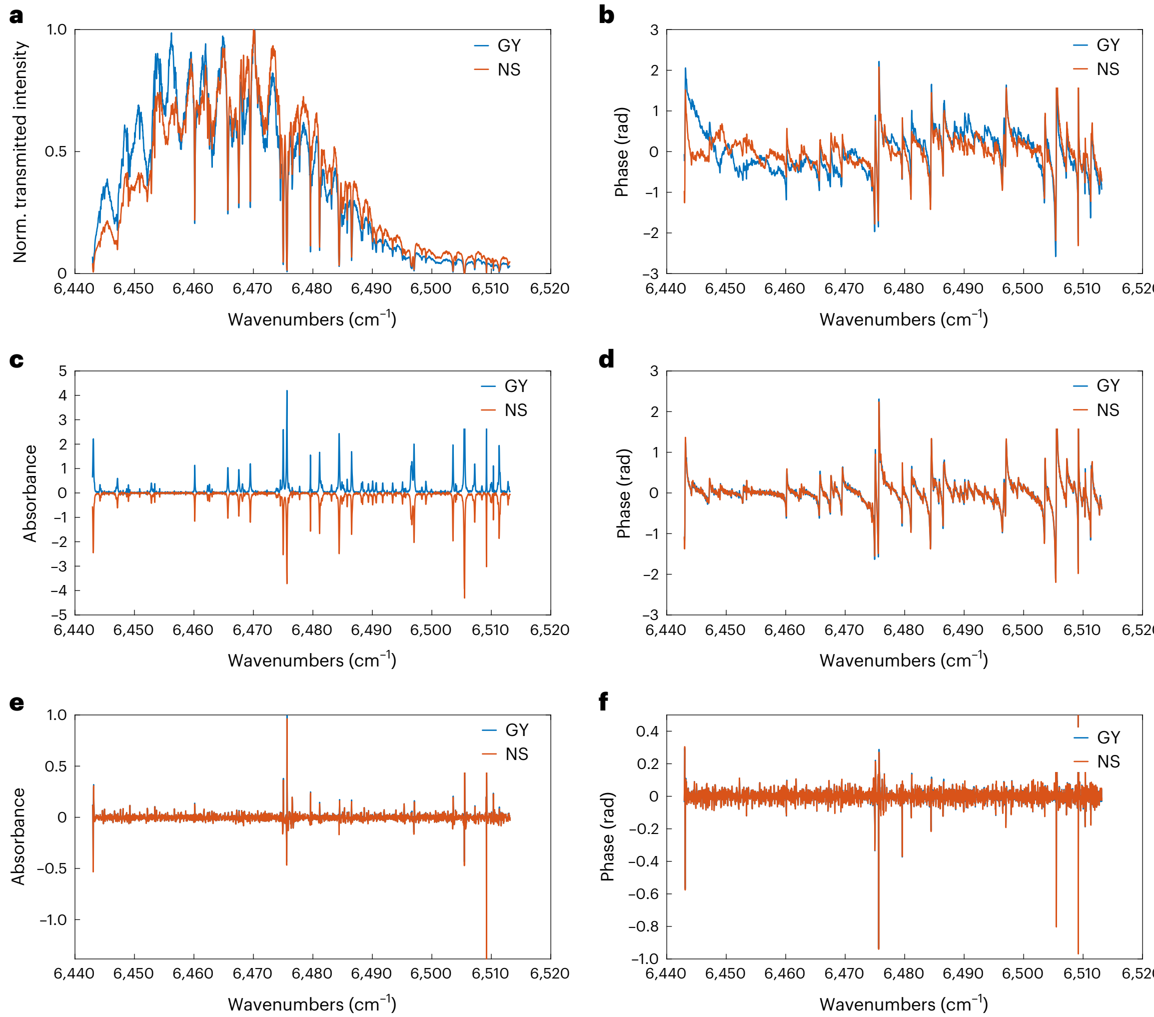

Fig. 3 | Original spectrum and resulting spectrum. Intensity spectrum and phase spectrum can be obtained at both Nanshan (NS) and Gaoyazi (GY). **a**, The normalized (Norm.) intensity spectrum. **b**, The phase spectrum after subtracting quadratic polynomials. **c**, The resulting absorbance. **d**, The resulting phase spectrum. **e**, The residual absorbance. **f**, The residual phase.

sharing a common frequency standard, and a mirror is placed on the other remote terminal as a reflector. Since the two combs are locked together, any absorption or dispersion will influence the interference of the two combs. DCS utilizes the change of interference to figure out the atmospheric property. In the past, there were two ways to achieve interference. One is that the dual combs interfere before passing through the atmosphere. The other is that one comb passes through the atmosphere first and then interferes with the local comb. The former can only provide the amplitude spectrum, while the latter can provide both amplitude and phase spectrum. However, both ways pass through the measured atmosphere twice, causing attenuation and turbulence twice. Moreover, the laser beam expands with distance and the mirror is usually not large enough to reflect all of the beam, causing additional geometric loss. For example, after passing through 100 km of turbulent atmosphere, it is normal for the beam to spread to a radius of 3 m, and the mirror of this size is expensive and impractical. This substantially limits their measurement range. Indeed, this issue turns out to be insoluble for satellite-ground links. Assuming a telescope aperture of 1 m and a mirror aperture of 0.5 m—a logical assumption considering the manufacturing process and satellite placement—the anticipated geometric attenuation reaches a substantial 105.8 dB for a geosynchronous orbit (GEO) spanning a satellite-ground distance of 36,000 km (Supplementary Information).

To increase the signal-to-noise ratio, a bistatic protocol was proposed[44]. In this protocol, the DCS transmitter and receiver are separated on the two ends of the path, and no reflector is needed. Here we introduce another bistatic protocol, as shown in Fig. 1b. The two combs are separated into two remote terminals. Both combs are sent to each other through the atmosphere and interfere with each other at the receiving terminals. With this configuration, both absorption and phase spectrum change by the atmosphere over the path can be achieved at each terminal. Since the comb passes through the atmosphere only once and no reflector is required, the link loss can be sharply reduced compared with the previous monostatic round-trip protocol.

However, this new protocol introduces a new problem. The basis of DCS is that the two combs need to share a common frequency standard. In the monostatic protocol, this condition is easy to achieve. Usually, a local frequency standard, such as a rubidium atomic clock, is exploited to lock both combs. In the bistatic protocol, each comb has its own frequency standard and the frequency difference and fluctuation of the two frequency standards will deteriorate the interference and reduce the frequency accuracy of the spectrum. Compared with the monostatic protocol, the frequency accuracy of the bistatic protocol is worse by a factor $M = f_r/\Delta f_r$, where f_r is the repetition rate of the local comb and Δf_r is the difference between the repetition rate of the two combs (see Methods for details). Figure 1c shows an example of spectra

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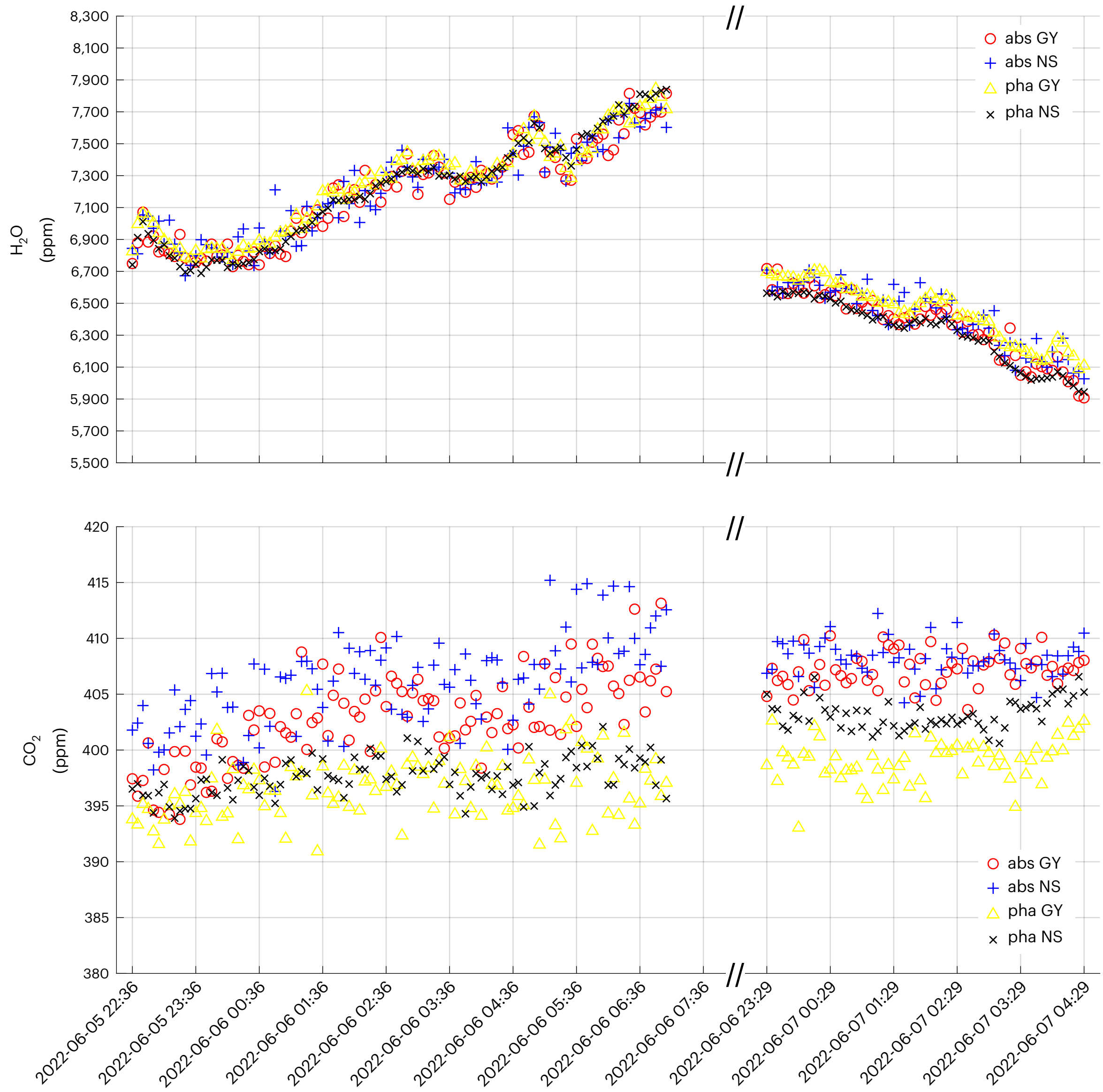

Fig. 4 | CO_2 concentration and H_2O concentration extracted independently from the absorbance and phase spectra at both terminals. The 'abs' refers to the concentration inverted from absorbance, and the 'pha' refers to the concentration inverted from phase spectrum. The changing trends of the four retrievals complied well.

with two frequency standards, where a frequency offset has occurred. Note that the offset is different for each measurement since the state of the frequency standard can change over time. To maintain the same frequency accuracy as the monostatic protocol, the frequency standard must be remotely locked. This problem can be naturally solved by the time–frequency dissemination or comparison technology[45,46,53–57]. Figure 1c also shows that the frequency offset is eliminated when the frequency standard is traced back to one terminal by frequency comparison. There are many ways to achieve frequency comparison. In our experiment, both frequency comparison and spectral information are obtained from the interferograms without the need for a separate link (see Methods for details).

Figure 2 illustrates our experimental set-up. Two terminals are located at Nanshan and Gaoyazi, Urumqi, Xinjiang Province, which are more than 113 km apart.

Each terminal has a rubidium clock that is used as the absolute frequency standard. Each comb is phase-locked to the local ultrastable laser (USL). The repetition rates f_r of the combs, used as the reference signals of the local sampling and reference clock of all electronics, are 250 MHz and 250 MHz + 2.5 kHz at Nanshan and Gaoyazi, respectively.

In terms of hardware, we develop high-power optical combs and ultrasensitive detection set-up to cope with the large loss of the link. The output power of each comb can reach 1 W after amplification in a two-stage high-power erbium-doped fibre amplifier (EDFA) with a 20 nm filtering spectrum bandwidth. The 3 dB bandwidth is approximately 7 nm centred at 1,545 nm.

Highly stable and efficient optical transmission systems, including polarization-maintaining fibres and dedicated optical telescopes, push the detected power into the nanowatt range. The optical paths before the telescope are fibre paths using polarization-maintaining fibres, most of which are integrated into an aluminium box. To maintain its stability, we controlled the temperature using a thermoelectric cooler, and the standard deviation of the temperature was 7 mK. The Global Positioning System (GPS) is used to synchronize the data collection start time, which is within 30 ns. Both improve the accuracy of the frequency comparison[57]. The power of the received signal swings widely owing to the characteristics of free space links over 100 km, so it is essential to determine when the signal is received and to observe the signal-to-noise ratio of the interferograms in real time. Here a threshold-triggered acquisition method is introduced: once the detector voltage exceeds a fixed threshold, an interferogram centred around the centerburst is captured. This method reduces many post-processing processes and, more importantly, reduces the pressure on memory and transmission bandwidth.

Two homemade optical telescopes with an aperture of 400 mm and a focal length of 1,600 mm are used to transmit and receive the laser. A direction-tracking system with a beacon laser and an adjustable mirror is developed to overcome large link loss at each telescope. Low noise-equivalent power balanced photodetectors are used to detect the interference between the local and received signals. All electronics are synchronized to the repetition rate of the local comb.

Results

In our experiment, the frequency accuracy can be improved by $M = f_r/\Delta f_r$, that is, 5 orders of magnitude, which is 1 GHz before and 10 kHz after the frequency comparison. To ensure that most of the phase unwrapping is correct, the return signal of the first interferogram is above 100 nW and is above 5 nW for other interferograms, equivalent to a total attenuation of 83 dB. Limited by data bandwidth, 40% of the entire single interferogram was captured, which reduced frequency resolution, and the frequency resolution is 605 MHz. The spectra span 6,000 comb teeth covering 6,505.5 cm^{-1} to 6,455.5 cm^{-1}. Figure 3a,b shows an example of the normalized intensity spectrum and the phase spectrum after subtraction of quadratic polynomials attributed to the air dispersion, acquired over 1 h, allowing long-term coherent averaging.

Molecular absorbance and phase signature scale linearly with concentration so that we can extract the gas concentration directly from the absorbance and the phase spectra separately. The reference spectrum cannot be obtained because the two combs cannot be placed on one terminal. A penalty least squares algorithm[58] (see Methods for details) is used to convert the absorbance or phase spectra to concentration. The model used for fitting is the High-Resolution Transmission Molecular Absorption Database (HITRAN) 2020, the complex Voigt line shape[59]. The path-averaged atmospheric pressure and temperature for the inversion are the averages of the sensors at both terminals. Figure 3c,d shows the resulting absorption spectrum and phase spectra. Figure 3e,f shows the residual absorbance and phase spectrum after fitting.

Figure 4 shows the CO_2 concentration and H_2O concentrations extracted independently from the absorbance and phase spectra at both terminals at 5 min intervals from 5 to 7 June. The changing trends of the four retrievals complied well. A slow increase in CO_2 concentration from evening to early morning was observed in the path that we measured with little human activity, which may be attributed to plant and microbial respiration or large-scale air movement and so on. The water vapour concentration was high on 5 June after the rainy day on 4 June and decreased substantially after the sunny day on 6 June. The CO_2 concentration from both absorbances agrees within ±10 ppm, 2 ppm on average, while the H_2O concentration agrees within ±300 ppm, 36 ppm on average. The sensitivity of the CO_2 concentration is calculated throughout the relative stability from 2022-06-06 23:29 to 2022-06-07 04:29, as shown in Fig. 5. The Allan variance of the concentration extracted from the absorbance was less than 2 ppm in 5 min and less than 0.6 ppm in 36 min.

The offset of the concentration extracted from the absorbance and phase spectra for both terminals occurs. We attribute this to two reasons. First, coarse atmospheric parameters introduce different offsets to the inversion of the intensity and phase spectra. Second, scattering in the optical path is unavoidable at high output power. The scattered light and the signal light will also interfere, influencing the inversion results (see Methods for details). Future work on retrieving atmospheric parameters with higher accuracy from atmospheric models or absorbance[36] and the cepstral-domain fitting technique[60] will make the inversion more accurate.

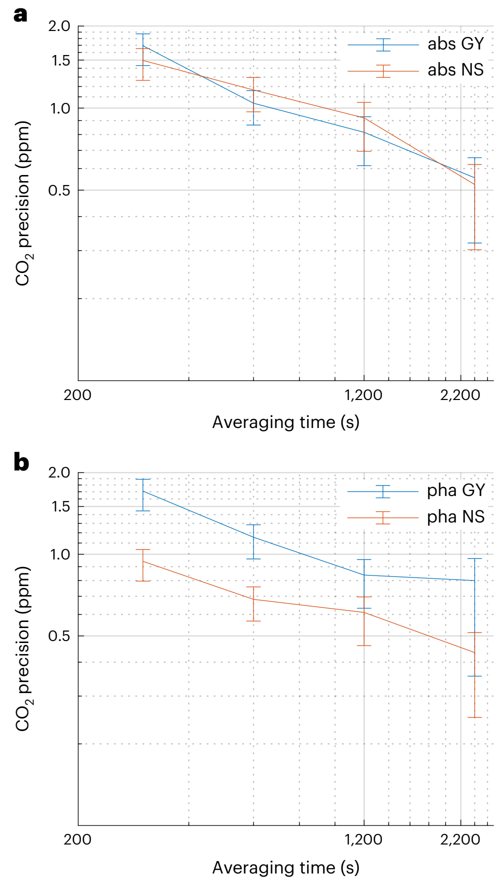

Fig. 5 | CO_2 precision (Allan deviation) for the four retrievals over a period of relative stability. The sensitivity of the CO_2 concentration is calculated throughout the relative stability from 2022-06-06 23:29 to 2022-06-07 04:29. The error bar is 1 σ. **a**, Allan deviation of concentration retrievals from absorbance. **b**, Allan deviation of concentration retrievals from the phase.

Conclusion

In summary, we demonstrate a bistatic DCS protocol combining time–frequency transfer technology. Several essential techniques have been used and verified, particularly time–frequency techniques, high-power stable optical frequency combs (OFCs) and high-efficiency transceiver telescopes. This protocol is more advantageous for long-range measurements. The full complex spectrum has been measured at both terminals over a 113 km open-air path, with a channel loss of 83 dB. The retrieved concentration precision of CO_2 is <2 ppm in 5 min and <0.6 ppm in 36 min in the intensity spectrum. This paves the way for satellite-ground DCS measurements with high resolution, broad spectrum and high sensitivity. Benefiting from the active high-accuracy GHG continuous monitoring ability of long-link DCS, it could become a worthy tool to calibrate the GHG satellites in the future. On the basis of our work, we anticipate that an ultra-long-range network of interlinked ground-space-based open-path DCS systems will play an important role in continuous and uninterrupted measurements of GHG fluxes. Our system requires plenty of ground stations to enlarge the coverage, which limits its global coverage. Combined with other existing satellites and technologies, some valuable data with unprecedentedly high precision can be acquired, which helps understand carbon sources and sinks and transport processes.

We developed high-power OFCs that helped us achieve horizontal 100 km measurements, but at the cost of a bandwidth of only 50 cm^{-1}. In the future, many new technologies such as soliton self-frequency shift[61] can be applied to OFCs to simultaneously develop high-power and large bandwidths to enable more GHG measurements such as CH_4. Besides, hardware can be optimized. The hardware should be upgraded to record the entire interferogram with a wider data bandwidth and transmit the data at a faster transfer rate. Furthermore, real-time phase correction method[62,63] would improve precision and time resolution.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41566-024-01525-9.

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Methods

The frequency accuracy of one frequency standard versus two frequency standards

When all phase-locked loops are closed for each comb, we have the relation $f_{\mathrm{CEO}} + N \cdot f_{\mathrm{r}} + f_{\mathrm{Beat}} = f_{\mathrm{CW}}$, where f_{CW} is the CW frequency of the USL, f_{r} is the repetition of the comb and N is the index of the tooth of comb locking to USL. In our experiment, the carrier-envelope offset frequency f_{CEO} and the beat frequency f_{Beat} are set to be equal but in opposite directions for each terminal so that $f_{\mathrm{CEO}} + f_{\mathrm{Beat}} = 0$, thus $N \cdot f_{\mathrm{r}} = f_{\mathrm{CW}}$. Extended Data Fig. 1 shows the locking direction of them at both terminals. Using the local rubidium clock as a reference, the frequency of the USL at each terminal is known by counting the repetition frequency of the comb.

Within the Nyquist sampling bandwidth, the beat frequency between two adjacent comb teeth is

$$\nu_{\mathrm{RF}} = 2f_{\mathrm{Beat}} - n \cdot \Delta f_{\mathrm{r}} - \Delta f_{\mathrm{CW}} \tag{1}$$

$\Delta f_{\mathrm{r}} = f_{\mathrm{r,GY}} - f_{\mathrm{r,NS}}$, $\Delta f_{\mathrm{CW}} = f_{\mathrm{CW,GY}} - f_{\mathrm{CW,NS}}$. n is an integer representing the position of the comb teeth. The complex spectrum corresponding to the intensity and phase of pairs of comb teeth reflects the atmosphere's response to the signal comb. So, the beat frequency, that is, radiofrequency, is needed to be converted to optical frequency. The optical frequencies corresponding to the nth pair of comb teeth are, respectively,

$$\nu_{\mathrm{OF,GY}} = f_{\mathrm{CW,GY}} - f_{\mathrm{Beat}} + n \cdot f_{\mathrm{r,GY}} \tag{2}$$

$$\nu_{\mathrm{OF,NS}} = f_{\mathrm{CW,NS}} + f_{\mathrm{Beat}} + n \cdot f_{\mathrm{r,NS}} \tag{3}$$

From equations (1), (2) and (3), we get

$$\nu_{\mathrm{OF,GY}} = (2f_{\mathrm{Beat}} - \nu_{\mathrm{RF}} - \Delta f_{\mathrm{CW}}) \cdot \frac{f_{\mathrm{r,GY}}}{\Delta f_{\mathrm{r}}} + f_{\mathrm{CW,GY}} - f_{\mathrm{Beat}} \tag{4}$$

$$\nu_{\mathrm{OF,NS}} = (2f_{\mathrm{Beat}} - \nu_{\mathrm{RF}} - \Delta f_{\mathrm{CW}}) \cdot \frac{f_{\mathrm{r,NS}}}{\Delta f_{\mathrm{r}}} + f_{\mathrm{CW,NS}} + f_{\mathrm{Beat}} \tag{5}$$

The absolute frequency of the USL at each terminal is calculated based on the local rubidium clock, and the frequency accuracy is 10 kHz. Therefore, the frequency accuracy of Δf_{CW} is the same order of magnitude. However, a factor of $M = \frac{f_{\mathrm{r}}}{\Delta f_{\mathrm{r}}}$ multiplied by Δf_{CW}, which is 100,000 in our experiment, greatly deteriorates the frequency accuracy. So to achieve the same frequency accuracy as the monostatic protocol, the accuracy of the frequency comparison must be less than $\frac{10\,\mathrm{kHz}}{100{,}000}$, that is, 0.1 Hz.

For the monostatic DCS protocol, where only one laser is used to lock two combs, Δf_{CW} is zero.

Frequency difference calculation and phase compensation

The electric field of the two combs at terminals A and B can be written as follows:

$$E_{\mathrm{A}}(t) = \exp(i2\pi\omega_{\mathrm{A}}t)\sum_n E_{\mathrm{A},n}\exp[i2\pi(nf_{\mathrm{r}}t + \varphi_{\mathrm{A},n})] \tag{6}$$

$$E_{\mathrm{B}}(t) = \exp(i2\pi\omega_{\mathrm{B}}t)\sum_n E_{\mathrm{B},n}\exp[i2\pi((nf_{\mathrm{r}} + \Delta f_{\mathrm{r}})t + \varphi_{\mathrm{B},n})] \tag{7}$$

Here ω_{A} and ω_{B} are the optical frequencies of one pair of most adjacent teeth of two combs. The difference between ω_{A} and ω_{B} is smaller than $f_{\mathrm{r}}/2$. After the low-pass filter with bandwidth $f_{\mathrm{r}}/2$, only the beat of most of the adjacent frequencies is maintained. There will be delay τ_0 when comb B is transmitted to comb A. Thus, the measured voltage at terminal A is as follows:

$$\begin{aligned} V(t,\tau_0) &\propto \mathrm{Im}\left[E_{\mathrm{A}}^*(t)\cdot E_{\mathrm{B}}(t-\tau_0)\right] \\ &= \mathrm{Im}\textstyle\sum_n \left(E_{\mathrm{A},n}^* E_{\mathrm{B},n} E_{\mathrm{atm},n}\right)\exp\{i2\pi\left[(\omega_{\mathrm{B}} - \omega_{\mathrm{A}} + n\Delta f_{\mathrm{r}})t\right. \\ &\quad + \left.\left[-n(f_{\mathrm{r}} + \Delta f_{\mathrm{r}})\tau_0 - \omega_{\mathrm{B}}\tau_0 + \varphi_{\mathrm{B},n} - \varphi_{\mathrm{A},n} + \varphi_{\mathrm{atm},n}\right]\right\} \end{aligned} \tag{8}$$

Here $E_{\mathrm{atm},n}$ is the atmospheric intensity response, and $\varphi_{\mathrm{atm},n}$ is the atmospheric phase response. We use $F(\nu,\tau)$ to represent the Fourier transform of the interferogram, and $\angle F(\nu,\tau)$ represents the spectral phase. Here ν is the frequency obtained from the fast Fourier transform of the interferogram. From equation (8), we get

$$\angle F(\nu,\tau_0) = -2\pi\tau_0\frac{f_{\mathrm{r}} + \Delta f_{\mathrm{r}}}{\Delta f_{\mathrm{r}}}[\nu - (\omega_{\mathrm{B}} - \omega_{\mathrm{A}})] - 2\pi\omega_{\mathrm{B}}\tau_0 + \varphi_{\mathrm{B}}(\nu) - \varphi_{\mathrm{A}}(\nu) + \varphi_{\mathrm{C}}(\nu) \tag{9}$$

At different times, the delays are different. Here we use τ to represent another delay:

$$\angle F(\nu,\tau) = -2\pi\tau\frac{f_{\mathrm{r}} + \Delta f_{\mathrm{r}}}{\Delta f_{\mathrm{r}}}[\nu - (\omega_{\mathrm{B}} - \omega_{\mathrm{A}})] - 2\pi\omega_{\mathrm{B}}\tau + \varphi_{\mathrm{B}}(\nu) - \varphi_{\mathrm{A}}(\nu) + \varphi_{\mathrm{C}}(\nu) \tag{10}$$

The difference between $\angle F(\nu,\tau)$ and $\angle F(\nu,\tau_0)$ are

$$\begin{aligned} \Delta\angle F(\nu,\tau) &= \angle F(\nu,\tau_0) - \angle F(\nu,\tau) \\ &= 2\pi\tfrac{f_{\mathrm{r}} + \Delta f_{\mathrm{r}}}{\Delta f_{\mathrm{r}}}[\nu - (\omega_{\mathrm{B}} - \omega_{\mathrm{A}})](\tau - \tau_0) + 2\pi\omega_{\mathrm{B}}(\tau - \tau_0) \end{aligned} \tag{11}$$

Make $\tau_0 = 0$, and the phase is a linear function to ν, and the slope $2\pi\frac{f_{\mathrm{r}}+\Delta f_{\mathrm{r}}}{\Delta f_{\mathrm{r}}}(\tau - \tau_0)$ is proportional to the delay τ. Therefore, τ can be calculated by fitting a line to $\Delta\angle F(\nu,\tau)$. Note that this delay is affected both by the drift between the frequency standards and by the atmosphere, so it varies with time.

The two-way method is usually used to counter the influence of the atmosphere. Let τ_{AB} and τ_{BA} represent the delay from A to B and the delay from B to A at the same moment, respectively. Then,

$$\begin{aligned} \tau_{\mathrm{AB}} &= \tau_{\mathrm{link}} - \tau_{\mathrm{clock}} \\ \tau_{\mathrm{BA}} &= \tau_{\mathrm{link}} + \tau_{\mathrm{clock}} \end{aligned} \tag{12}$$

So,

$$\tau_{\mathrm{clock}} = \frac{1}{2}(\tau_{\mathrm{BA}} - \tau_{\mathrm{AB}}) \tag{13}$$

where τ_{link} is the delay of the atmosphere and τ_{clock} is the time fluctuations of frequency standards. The fractional frequency deviation is the derivative of τ_{clock} and frequency comparison can be calculated in real time. As a by-product of this frequency comparison method, the optical path change owing to turbulence can also be calculated as

$$\tau_{\mathrm{link}} = \frac{1}{2}(\tau_{\mathrm{BA}} + \tau_{\mathrm{AB}}) \tag{14}$$

From this, the phase noise can be analysed (Supplementary Information).

The signal-to-noise ratio of a signal interferogram is too low to observe the spectral information. From equation (11), we can get

$$\angle F(\nu,0) = \angle F(\nu,\tau) + \Delta\angle F(\nu,\tau) \tag{15}$$

A linear fitting of ν represents $\Delta\angle F(\nu,\tau)$. All interferograms used for calculation have the same delay as the first interferogram after adding $\Delta\angle F(\nu,\tau)$, and phase compensation is achieved.

Protocol comparison of sending two combs at one terminal versus sending two combs to each other at different terminals

Note for the previous bistatic protocol[44] that sends out both combs through the atmosphere and acquires data at the end of the path, frequency comparison is not required to maintain frequency accuracy. However, some common time base is still needed to synchronize the acquisition electronics with the frequency comb repetition rate for coherent averaging. In this protocol, the phase information is missing. Therefore, only the intensity spectrum could be measured. By contrast, the comb is sensitive to phase in our protocol. So, the intensity spectrum and phase spectrum can be obtained at each terminal, and the phase noise of turbulence can also be analysed. Besides, multi-mode fibre could be used to couple to photodetector in the phase-insensitive protocol, which means the greater coupling efficiency. However, in the case of ground-satellite or 100 km horizontal links, the loss is so large that the received power is only a few nanowatts on average for each comb. But in our configuration, the local power is always stable and sufficient at the received terminal, so the signal-to-noise ratio for a single interferogram is better in most cases in comparison. In addition, balanced photodetector cannot be used in the phase-insensitive protocol, which means a loss of 3 dB in comparison.

Detailed experimental set-up and parameters

The experimental set-up of one terminal is shown in Extended Data Fig. 2. The set-ups of the two terminals are identical. Each terminal has a rubidium clock (SRS FS725, 10 MHz) that acts as the absolute frequency standard for all locking electronics. Each comb is phase-locked to the local USL (Stable Laser Systems SLS-INT-1550-1) with sub-1 Hz linewidth at the optical frequency of 193.4 THz, while the carrier-envelope offset frequency f_{CEO} is locked by way of the f–$2f$ method. Both frequency instabilities are reached 1×10^{-21} at 10,000 s. The carrier-envelope offset (CEO) frequency (f_{CEO}) and the beat frequency (f_{Beat}) are set to be equal and in opposite directions so that they cancel each other out. The frequency of the USL used to lock the OFC can be known by recording the repetition rate of the OFC.

The OFC is amplified up to 1 W. Note that the vertical atmospheric scale height of the well-mixed gas is about 8 km, which is a factor of 10 smaller than the 100 km in our experiments. So, the absorption signal in our experiment is much stronger than a GEO configuration. If a wavelength range with a strong scattering cross section was used, as in other experiments, the CO_2 absorption would be completely saturated, making it unsuitable for atmospheric absorption measurements over hundreds of kilometres. To avoid damage caused by high power density, a chirped pulse with 60–100 ps is produced. Water chillers are used to avoid damage to the EDFAs owing to overheating. The loss for a satellite-ground link is different in downlink and uplink, which is about 54.9 dB for downlink and 77.1 dB for uplink (Supplementary Information). The downlink loss is much less, so the laser placed on the satellite does not need to have an output power of 1 W. A few tens of milliwatts would be sufficient.

A 1 × 3 beam splitter divides the beam into three parts with a ratio of 1:2:97. The 1% part is used as the local beam to interfere with a signal. The 97% part is connected to the telescope. The last 2% part goes through a 0.8 nm filter and then beats with the USL. Interference, locking and transmission are all integrated in an aluminium box with temperature control to stabilize the temperature. The clock for recording and processing signals is synchronized with the repetition rate of the local OFC. A GPS is used to synchronize the start time of both terminals.

Low noise-equivalent power balanced photodetectors (New Focus Model 1817) are used to detect the interference between the local and received signals. Some of the noise was filtered out using a low-pass filter with a bandwidth of 100 MHz. The 14-bit analogue-to-digital converters (ADCs) are used to digitize.

In our experiments, the acquisition time for a single interferogram is 0.5 ms, but the dead time for transmitting and storing the data is 9 ms. So actually, a lot of useful interferograms are lost, which may lead to lower precision. For a DCS system, with f_r fixed, the wider the optical bandwidth, the smaller the Δf_r becomes and the more points one interferogram has. In the case of a limited data bandwidth, if one wants to cover a wider optical bandwidth, the spectral resolution becomes worse, since the interferograms are recorded in even smaller proportions. A total of 1,024 points around the interferogram centerburst are used for Fourier transform to extract the smoothed phase spectrum, which is used for phase compensation and frequency comparison.

Optical telescope and acquiring, pointing and tracking system

Extended Data Fig. 3 shows the detailed set-up of the telescope. The primary mirror of each transceiver telescope at each terminal is a Cassegrain reflector with an aperture of 400 mm and a focal length of 1,600 mm. To simulate the different transmit and receive paths of a satellite-ground link where the relative angular motion separates the backwards and forwards transfers, emitted and received OFCs are separated using horizontal and vertical polarization. As shown in Extended Data Fig. 3, red lines and blue lines represent different polarization. A wavelength division multiplexer (WDM) at 20 nm is used to filter received signal light. A beacon laser for direction tracking at wavelengths of 785 nm and 914 nm, respectively, is placed on the telescope at each terminal. To maintain a stable link channel over a long period of time, a two-stage ATP system called coarse tracking and fine tracking, respectively, is utilized. For coarse tracking, the beacon laser is detected directly by CMOS1 at the other terminal and feedback to the two-dimensional rotating platform used to support the telescope in both azimuth and pitching directions. Coarse tracking is used to resist slow but violent excursions owing to mechanical stress relief and ground settlement. For fine tracking, the beacon laser is retracted into the telescope, separated from the signal OFC by a dichroic mirror (DM), and then detected by the CMOS2. A fast steering mirror (FSM) driven by piezo ceramics (PI) is feedback to restrain the beam quiver caused by atmospheric turbulence. Since the optical comb signal and the fine-tracking beacon light are co-axial, the quality of the fine tracking directly affects the receiving efficiency. The closed-loop bandwidth of the fine tracking is more than 150 Hz, which is sufficient to resist atmospheric turbulence.

Penalized least squares algorithm

The measured complex spectral signal $s(\nu)$ can be written as follows:

$$s(\nu) = b \times a + n \tag{16}$$

where $s(\nu)$ is the measured complex spectral signal, ν is the optical frequency, b is the laser spectrum, which is relatively smooth, a is the molecular resonances, and n is the frequency domain noise.

$$a = \exp\left(-\sum_{i=1}^{q} \alpha_i c_i L\right) \tag{17}$$

where q is the number of gas species, which in our experiment is 2; α is the complex absorption rate of 1 ppm molecule; c is path-averaged mole fractions; and L is the path length. Ignoring noise, taking logarithms on both sides of the above and making $\sigma = \log(s)$, $\beta = \log(b)$, we get

$$\sigma = \beta - \sum_{i=1}^{q} \alpha_i \cdot c_i \cdot L \tag{18}$$

Writing in a matrix form

$$\begin{bmatrix} I & -\alpha_1 & -\alpha_2 \end{bmatrix} \begin{bmatrix} \beta \\ c_1 \cdot L \\ c_2 \cdot L \end{bmatrix} = \sigma \tag{19}$$

and making $\left[\,I\ -\alpha_1\ -\alpha_2\,\right] = G, \begin{bmatrix} \beta \\ c_1 \cdot L \\ c_2 \cdot L \end{bmatrix} = m$, then we get

$$G \cdot m = \sigma \tag{20}$$

After the Fourier transform of an interferogram with 41,350 points, limited by the spectrum range, the effective numbers are only 2,567 points. Therefore, σ is a 2,567 × 1 column vector. I is a 2,567 × 2,567 unit matrix. G is a 2,567 × (2,567 + 2) matrix. m is a (2,567 + 2) × 1 column vector. The above equation is an underdetermined equation with infinitely many solutions. It is necessary to add regularization constraints to determine the unique solution. The fidelity of σ against $G \cdot m$ can be expressed by the population variance between them:

$$F = \sum_{i=1}^{r} \left(\sigma_i - (Gm)_i\right)^2 = \|\sigma - G \cdot m\|^2 \tag{21}$$

The roughness of fitted data β can be expressed by the sum of squares of its difference:

$$R = \left\|\boldsymbol{D} \cdot \beta\right\|^2 \tag{22}$$

where r is the dimension of the vector, that is, 2,567 here, and $\boldsymbol{D}$ is the difference matrix. According to ref. 58, the difference order is usually set to 2, that is,

$$\boldsymbol{D} = \begin{bmatrix} 1 & -1 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & -1 & 0 \\ 0 & \cdots & 0 & 1 & -1 \end{bmatrix} \tag{23}$$

The dimensions of the matrix D are (2,567 − 1) × 2,567. Therefore, $R = \sum_{i=2}^{r} \left(\beta_i - \beta_{i-1}\right)^2$.

The objective function Q is the sum of the fidelity F and the roughness R with different weights:

$$Q = F + \lambda \cdot R = \|\sigma - G \cdot m\|^2 + \lambda \cdot \left\|\boldsymbol{D} \cdot \beta\right\|^2 \tag{24}$$

where λ is the weighting factor. Letting $\boldsymbol{D}_2 = \left[\,\boldsymbol{D}\ \boldsymbol{0}\,\right]$, we get

$$Q = \|\sigma - G \cdot m\|^2 + \lambda \cdot \left|\left|\boldsymbol{D}_2 \cdot m\right|\right|^2 \tag{25}$$

Here 0 is a 2,567 × 2 matrix of zeros to ensure that the differential is only realized at β. The inversion of concentration and background spectrum can be written as an optimization problem with constraints. The optimization function is a convex function with a minimum value. At the minimum, the derivative is 0. We get

$$\frac{\partial Q}{\partial m} = 0 \tag{26}$$

$$m = \left(G^T G + \lambda \cdot \boldsymbol{D}_2^T \boldsymbol{D}_2\right)^{-1} \cdot G^T \cdot \sigma = \boldsymbol{H} \cdot \sigma \tag{27}$$

where $\boldsymbol{H} = \left(G^T G + \lambda \cdot \boldsymbol{D}_2^T \boldsymbol{D}_2\right)^{-1} \cdot G^T$, and the weighting factor λ depends on the smoothness of the substrate, that is, β. The smoother the substrate, the greater the weight factor. From this inversion method, the baseline and concentration could be acquired at the same time.

Deviation analysis of reflected stray light

The output power is so high that nonlinear effect and scattering are inevitably introduced. Supposing that there exists an interferogram that is γ times as strong as the original with a delay Γ, the time domain signal becomes

$$S'(t) = S(t) + \gamma S(t - \Gamma) \tag{28}$$

Then the measured complex spectral signal becomes

$$s'(\nu) = \left(1 + \gamma e^{i2\pi\Gamma\nu}\right) s(\nu) \tag{29}$$

The logarithmic spectrum is

$$\sigma'(\nu) \approx \sigma(\nu) + \log\left(1 + \gamma e^{i2\pi\Gamma\nu}\right) \tag{30}$$

When $\gamma \ll 1$, $\log(1 + \gamma e^{i2\pi\Gamma\nu}) \approx \gamma e^{i2\pi\Gamma\nu}$, so

$$\sigma'(\nu) = \sigma(\nu) + \gamma e^{i2\pi\Gamma\nu} \tag{31}$$

Therefore, a cosine modulation $\gamma\cos(\pi\Gamma\nu)$ is introduced in the intensity spectrum and a sinusoidal modulation $\gamma\sin(\pi\Gamma\nu)$ is introduced in the phase spectrum.

Data availability

All data generated or analysed during this study are included in this article (and its Supplementary Information files). All data are available from the corresponding author upon reasonable request. Source data are provided with this paper.

Code availability

All relevant codes or algorithms are available from the corresponding author upon reasonable request.

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Author contributions

H.-F.J., X.-H.X., Q.Z., X.-K.D. and J.-W.P. conceived the experiment. J.-J.H., W.Z., R.-C.Z., Q.S., J.-Y.G., H.-F.J., X.-H.X. and Q.Z. designed the dual-comb spectroscopy set-up. J.-G.R., T.Z. and J.-J.J. built the optical telescopes. L.H., X.-X.P. and H.-F.J. developed the optical frequency combs and amplifiers. Q.S., M.L., J.-Y.G. and J.-J.H. developed the linear optical sampling optics and electrics. X.-P.S., M.L., Q.S. and J.-Y.G. designed the data acquisition software. J.-J.H., W.Z., R.-C.Z., Q.Y., J.-Y.G., Q.S. and M.L. developed the data process algorithms and designed the data process software. All authors carried out the experiment, analysed the data and contributed to the writing of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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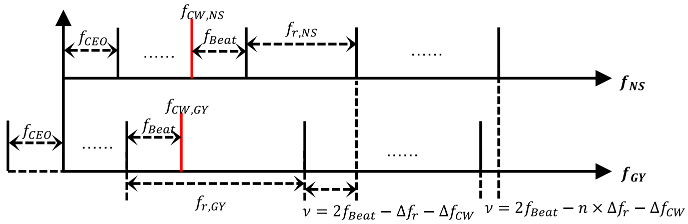

Extended Data Fig. 1 | The locking direction of the frequency combs at both terminals. The f_{CEO} and the f_{Beat} are set to be equal but in opposite directions for each terminal.

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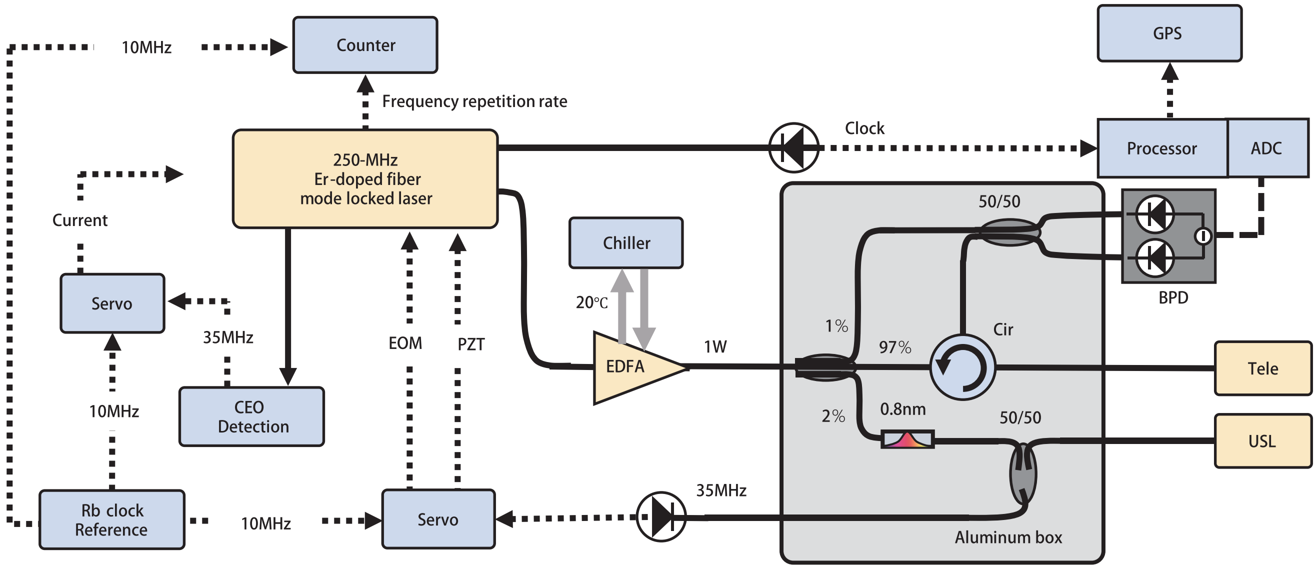
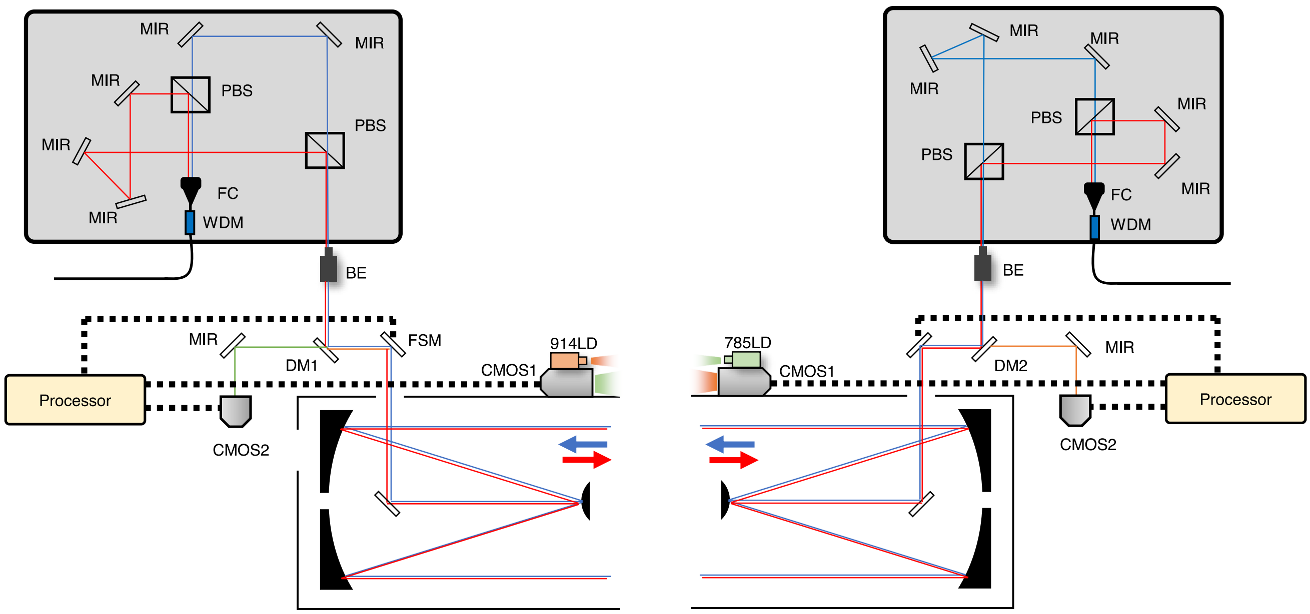


Extended Data Fig. 2 | Detailed experimental optical setup of single terminal. USL: Ultra-stable laser; Tele: Telescope; BPD: Balanced photondetector; GPS: Global Position System; Cir: Circulator; EDFA: Erbium-doped fiber amplifiers.



Extended Data Fig. 3 | Detailed experimental optical setup of Telescope. MIR: Mirror; PBS: Polarizing beam splitter; FC: Fiber collimator; WDM: wavelength division multiplexer; BE: Beam expander; FSM: Fast steering mirror; LD: Laser diode; DM1: Dichroic beam splitter, 785nm transmitted, 1545nm reflected; DM2: Dichroic beam splitter, 914nm transmitted, 1545nm reflected; CMOS: Complementary metal–oxide semiconductor.